Robert Koch devised the solid medium (by using gelatin) to grow and isolate the microorganisms.

5.1 Significance of Culturing Microorganisms

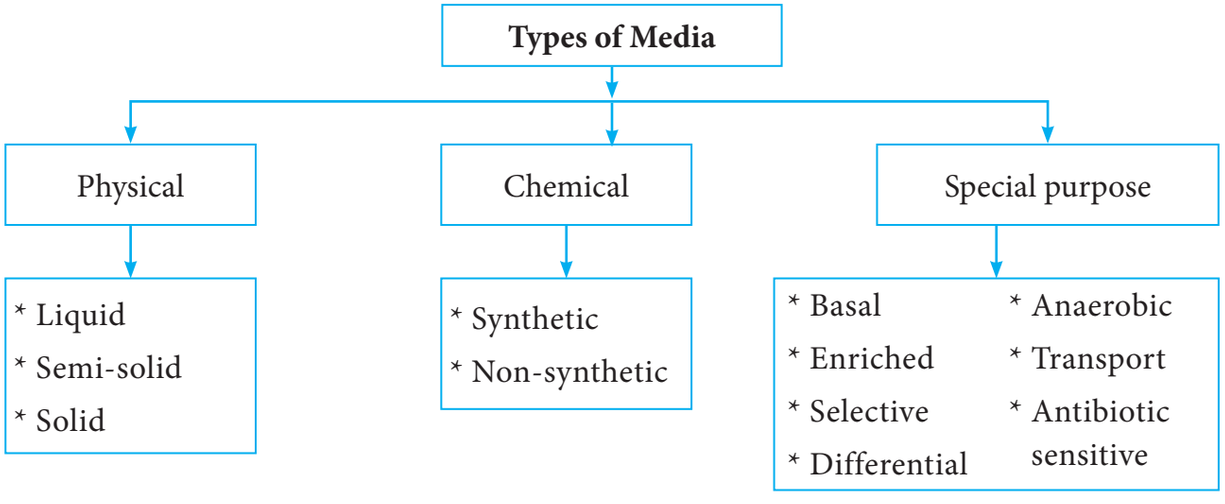
- To isolate microorganisms from any samples
- To study the morphology and biochemical characteristics of microorganisms
- To maintain the stock culture
- To identify disease causing microorganisms
- To study the role of microorganisms in the production of industrially important products

5.2 Bacteriological Media and its Types

Generally microorganisms occur as mixed culture in nature. Human beings, animal bodies and other natural resources harbour microbes in mixed population. By using appropriate media, microorganisms can be grown separately in pure form and can be studied. For successful cultivation of a given microorganism, it is necessary to understand the nutritional requirements of that microorganism and then supply the essential nutrients in proper form and proportions in culture medium. Flowchart 5.1 describes the types of media. A common bacteriological medium has Carbon and Nitrogen sources along with buffering agents. Most of the media are prepared using dehydrated components. The basic components are peptone, beef extract, meat extract, yeast extract and agar (Table 5.1).

Table 5.1: Common ingredients of a culture media

S.No	Ingredients	Source of
a.	Peptone (protein hydrolysates)	Carbon, nitrogen, energy
b.	Beef extract (Extract of lean beef)	Aminoacids, vitamins, minerals
c.	Yeast extract (Brewer's yeast)	Vitamin B, Carbon, Nitrogen
d.	Agar	Solidifying agent



Flowchart 5.1: Types of media

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